MILITARY SPECIFICATION

DRAPE, OPERATING MICROSCOPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable, transparent, plastic drapes for covering Zeiss-type, binocular, surgical operating microscopes.

1.2 Classification. The drapes shall be of the following classes, as specified (6.2):

Class I - With two eyepieces.
Class II - With four eyepices.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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MILITARY

MIL-P-116 - Preservation-packaging, Methods of.
MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING & MATERIALS (ASTM)

ASTM D-882 - Tensile Properties of Thin Plastic Sheeting.
ASTM D-1003 - Haze and Luminous Transmittance of Transparent Plastics.
ASTM D-1894 - Coefficient of Friction of Plastic Film and Sheeting.
ASTM D-1922 - Propagation Tear Resistance of Plastic Film and Thin Sheeting.
ASTM E-96 - Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 56A Standard for the Use of Inhalation Anesthetics

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Materials. All materials used in the construction of the drapes shall be suitable, in all respects, for the purpose intended.

3.2.1 Plastic film. Polypropylene film used in the body of the drape shall have the properties listed in table I when tested as specified in 4.4.1.

TABLE I. Properties of polypropylene film.

Characteristic	Units	Value
Tensile strength	pounds/in.^{2}	
Machine direction		7,000 min.
Transverse direction		4,000 min.
Elongation	percent	
Machine direction		650 min.
Transverse direction		720 min.
Tear strength	gms./.001 in.	
Machine direction		20 min.
Cross direction		100 min.
Moisture vapor transmission rate	gms/100 in^2/ 24 hrs./.001 in. at 37.8° C	0.7 ± 10%
Coefficient of friction	number	0.2 max.

3.2.2 Plastic lens cover. The clear plastic material which covers the lens shall be free of distortion and shall have the properties listed in table II when tested as specified in 4.4.3.

TABLE II. Properties of plastic lens cover.

Characteristic	Units	Value
Luminous Transmittance	percent	90 min.
Haze	percent	1 max.

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3.3 Design and construction. Drape shall be constructed of polypropylene film, at least .0009 inch thick, conforming to 3.2.1. The drape shall cover a Zeiss-type surgical operating microscope and supporting horizontal arm, permitting access to all controls while protecting the surgeon and his staff from contamination. All seams in the body of the drape shall be heat sealed. Seams, and the attachment of the lens ring, shall be continuous to maintain sterility. One rubber band, suitable for attaching the drape to the eyepiece housing and conveniently located for that purpose, shall be supplied for each eyepiece cover.

3.3.1 Lens ring. A lens ring shall be secured to the drape by means of a suitable adhesive or by heat sealing. The lens ring shall contain a removable, optically-clear, distortion-free plastic lens cover conforming to 3.2.2. The lens ring shall fit snugly over an objective-lens housing having an outside diameter of 1.89 ± 0.03 inches (in.) (48.0 ± 0.8 millimeters (mm.)) and a minimum length of 0.2 in. (5.1 mm.). When tested as specified in 4.4.4, the lens ring shall sustain a weight of 1/2 pound without being dislodged from the lens housing.

3.3.2 Class I drape. The class I drape shall be in conformance with figure 1 and as specified herein.

3.3.3 Class II drape. The class II drape shall be in conformance with figure 2 and as specified herein.

3.4 Sterility. The drapes shall be sterile when tested with a biological indicator as specified in 4.4.5.

3.4.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include.

a. Item Identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, or inoculated product.
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of a laboratory or a designated qualified alternate.

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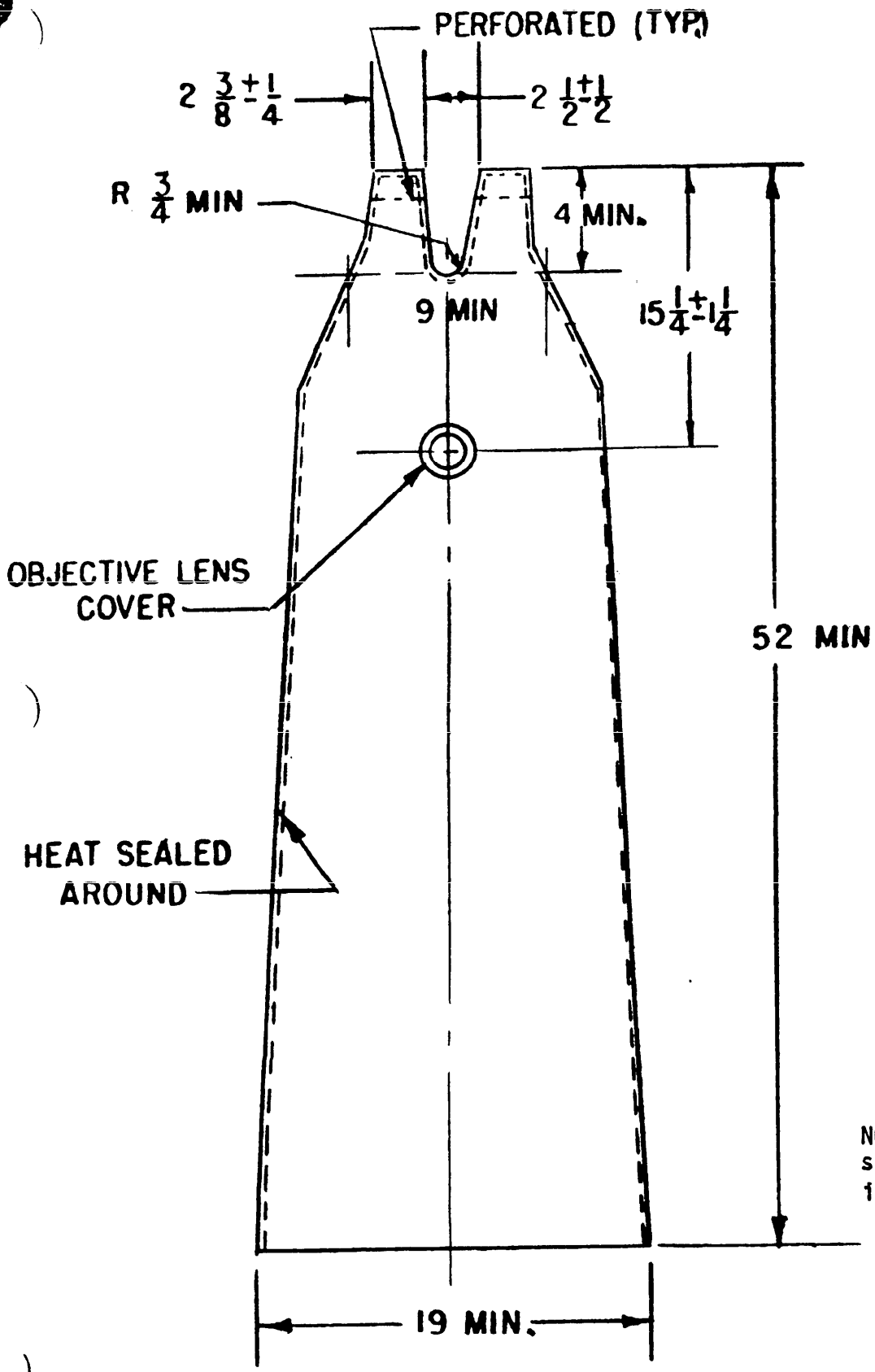


FIGURE 1. Drape, Operating Microscope, with Two Eyepieces.

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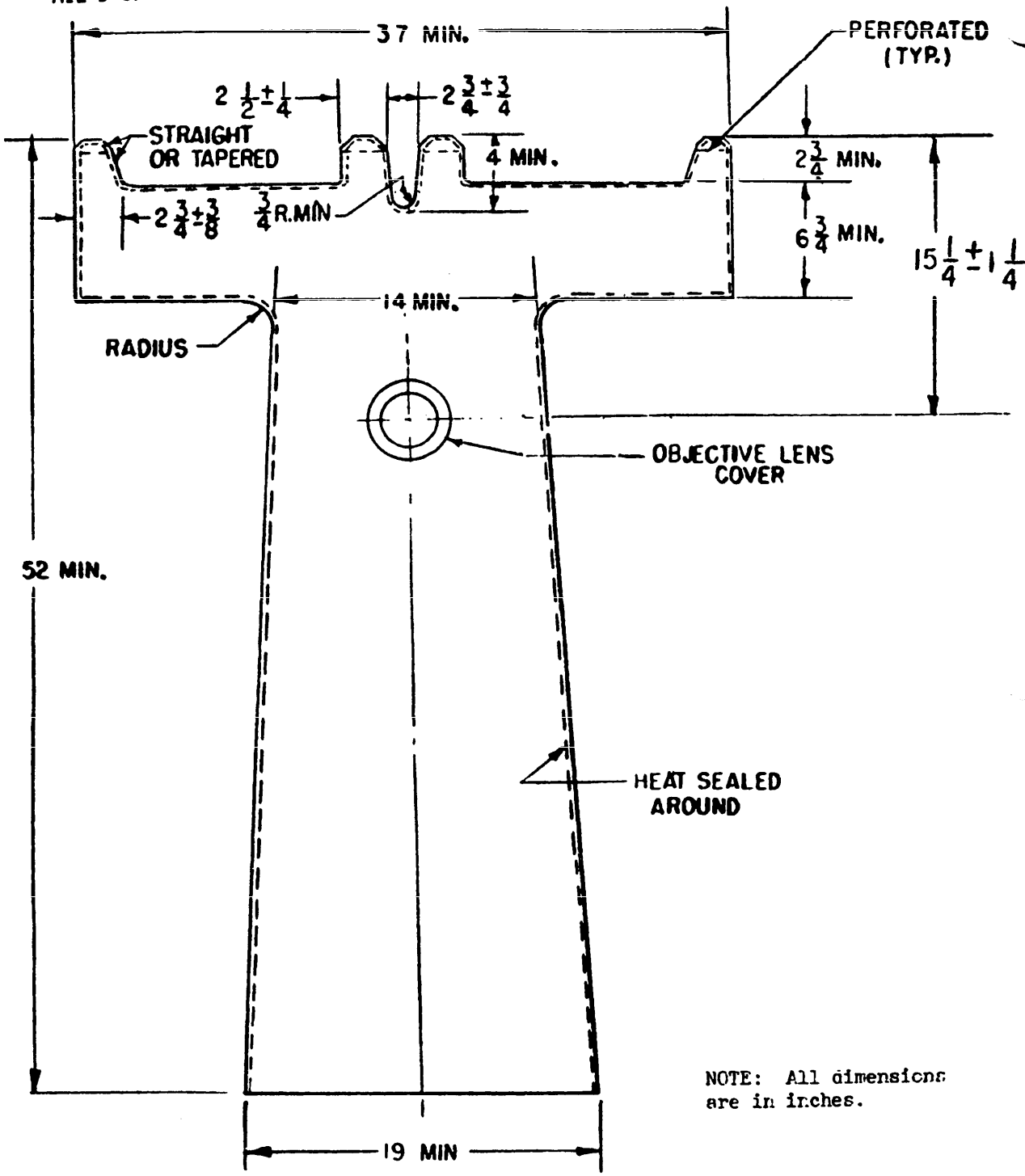


FIGURE 2. Drape, Operating Microscope, with Four Eyepieces.

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When contract provides for acceptance at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When contract provides for acceptance at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.4.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.5 Surface resistivity or static decay. When tested as specified in 4.4.2, the surface resistivity of the body of the drape shall not exceed 1×10^{11} ohms per square inch or, if the static decay test is employed, the time for the indicated specimen potential to drop to 10 percent of its maximum value shall not exceed 1/2 second.

3.6 Workmanship. Workmanship shall be first class throughout. Drapes shall be free from defects which detract from their appearance or impair their serviceability. Lenses and lens rings shall be free of sharpness and flash. The body of the drape shall contain no tears or holes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on drapes when a first article sample is required (see 3.1). This inspection shall include examination of 4.3. and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table III. The unit of product for sampling purposes shall be one drape.

TABLE III. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	S-4	1.0
Total defects (Major and Minor combined)	S-4	4.0
For dimensional examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table IV for an inspection level of S-2 for all tests except sterility, which shall be S-3. The acceptance level shall be zero for all sample sizes.

TABLE IV. Sampling for tests.

Component (Unit of product)	Characteristic	Requirement	Test procedure
Plastic film	Tensile strength	3.2.1	4.4.1
	Elongation	3.2.1	4.4.1
	Tear strength	3.2.1	4.4.1
	Moisture vapor transmission rate	3.2.1	4.4.1
	Coefficient of friction	3.2.1	4.4.1
	Surface resistivity or static decay	3.5	4.4.2
Lens cover	Luminous transmittance	3.2.2	4.4.3
	Haze	3.2.2	4.4.3
Lens ring	Adherence	3.3.1	4.4.4
Drape	Sterility	3.4	4.4.5

4.3 Examination. The drapes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table V. Examination shall not be restricted to the classified possible defects listed below.

TABLE V. Classification of defects.

Major	
101	Drape body has tears or holes.
102	Any seam not continuously sealed.
103	Lens ring not securely or continuously sealed to drape.
104	Lens cover does not fit snugly into lens ring.
105	Object viewed through lens cover is unclear or distorted.
106	Lens cover cannot be removed without damage to lens ring or body of drape.
107	Lens cover has chip, crack, or visual imperfection.
108	Rubber band(s) missing.

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TABLE V. Classification of defects. (Continued)

Minor	
201	Any part of drape has dirt, oil, grease, or other foreign matter.
202	Instructions missing, incomplete, or illegible.
203	Identification marking not as specified.

4.3.2 Dimensional examination. The drapes shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be as classified as a defect.

4.3.3 Packaging inspection. The inspection of the individual package shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Leakage test in group B is not applicable. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Plastic film. Plastic film shall be tested in accordance with ASTM Standard Methods of Test listed in table VI.

TABLE VI. Testing of plastic film.

Characteristic	Test method
Tensile strength	D-882
Elongation	D-882
Tear strength	D-1922
Moisture vapor transmission rate	E-96
Coefficient of friction	D-1894

4.4.2 Surface resistivity or static decay. Surface resistivity or static decay test shall be conducted in accordance with NFPA 56A.

4.4.3 Lens-cover. The plastic lens-cover shall be tested in accordance with the ASTM Standard Methods of Test listed in table VII.

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TABLE VII. Testing of lens-cover.

Characteristic	Test method
Luminous transmittance	D-1003
Haze	D-1003

4.4.4 Lens ring adherence. The lens ring shall be fitted in place over the objective-lens housing of an operating microscope and the lens shall be directed straight down. (The housing shall have a maximum diameter of 1.92 in. (48.8 mm.).) A length of string or wire shall be affixed to one point on the periphery of the lens ring. A 1/2 pound weight shall be attached to the free end of the string or wire and left suspended for 5 seconds.

4.4.5 Sterility test. The drapes shall be tested for sterility in accordance with the method specified in the USP using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.5.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for type I sterilization and 30 for type II sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table IV, Sampling for tests.

4.4.5.1 Inoculated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
I Steam	B. stearothermophilus	55° - 65° C
II Dry Heat	B. stearothermophilus, or	55° - 65° C
	B. subtilis var. niger (or globigii)	30° - 35° C
Gas	B. subtilis var. niger (or globigii)	30° - 35° C
Radiation	B. pumilus	30° - 35° C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.5.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.5 Sterility Test and 4.4.5.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.5.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.5.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.5.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

5. PACKAGING

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing twelve (12) drapes of class I or class II as specified (see 6.2) constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Individual package. Each drape shall be neatly folded and packaged in a sealed paper or plastic-peel-open type envelope which will maintain sterility of contents unless damaged or opened. The envelopes shall be so designed as to permit dispensing of contents by sterile techniques.

5.1.1.1.2 Unit package. Twelve (12) individual packages shall be packaged in a box of appropriate size and design. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Drapes shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A. Four (4) units of issue shall be packed in an exterior container constructed in accordance with PPP-B-636, class weather resistant, grade V3c. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Four (4) units of issue shall be packed in an exterior container constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. The packaged drapes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each individual package shall be marked with the contract or purchase order number or with the name or registered trademark of the manufacturer, the lot (control) number, requirements of the Federal, Food, Drug and Cosmetic Act, and the legends: "STERILE" and "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" and instructions for application.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Lot (control) number and name of contractor shall be shown. The method of sterilization (G-GAS, R-RADIATION, S-STEAM, C-COLD, D-DRY HEAT STERILIZATION) shall be shown following the level of packaging and date of packaging. Example: A-4/77-G. Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and method of sterilization as specified for the unit package shall be shown.

6. NOTES

6.1 Intended use. The drapes are intended to cover operating microscopes to maintain sterility during operations in military medical facilities.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Class required (see 1.2).
d. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulation. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification Designation Class
6530-00-419-4556	DRAPE, OPERATING MICROSCOPE, Transparent Plastic, with Two Eyepieces, 12 s.	I
6530-00-420-0513	DRAPE, OPERATING MICROSCOPE, Transparent Plastic, with Four Eyepieces, 12 s.	II

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Project No. 6530-1420

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-37587 DRAPE,OPERATING MICROSCOPE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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